ShopSmart Retail Analytics Project Procedure

1. Project Overview

This project was conducted to analyze ShopSmart Retail's sales performance, product return rates, and inventory management. The objective was to transform raw transactional data into actionable business insights through data cleaning, feature engineering, exploratory data analysis, and dashboard development in Microsoft Excel.

2. Data Sources

The project utilized the following datasets:

- Customers.csv
- Products.csv
- Orders.csv
- Stores.csv
- Returns.csv
- Employees.csv
- Suppliers.csv
- Inventory.csv
- Order_Items.csv

These datasets were imported into separate worksheets in a single Excel workbook for analysis.

3. Data Cleaning

The following data-cleaning procedures were performed:

Formatting

- Formatted date fields were as dates (dd/mm/yyyy).
- Ensured all currency fields were formatted in Naira.
- Verified quantity fields were stored as numeric values.
- Used TRIM() function to remove trailing spaces from text data
- Used CONCAT() to join the first name and last name columns in the customers table.

Missing Values

- Identified blank and missing values and replaced them with N/A or – where appropriate.

Duplicate Records

- Checked for duplicate Order IDs.
- Removed duplicate records using the “Remove duplicates” button

4. Data Integration

The datasets were integrated using the primary and foreign keys to establish relationships between tables:

Table 1: Shows the primary key, foreign keys, and relatonships between tables

Table	Primary Key	Foreign Keys	Relationships
Suppliers	Supplier_ID	None	NONE
Stores	Store_ID	None	Orders Inventory Employees
Returns	Return_ID	Order_ID	Orders
Products	Product_ID	None	Orders Order Items Inventory
Orders	Order_ID	Customer_ID Product_ID Store_ID	Customers Products Stores
Order Items	Order_Item_ID	Order_ID Product_ID	Orders Products
Inventory	Inventory_ID	Product_ID Store_ID	Products Stores
Employees	Employee_ID	Store_ID	Stores
Customer	Customer_ID	None	Orders

XLOOKUP was used to bring the following columns from other tables into the main Orders table:

For example:

=XLOOKUP([@[Customer_ID]],ShopSmart_Customers[Customer_ID],ShopSmart_Customers[Customer_Name])

- Refund_Amount
- Customer_Name
- Gender
- Store_Name
- Store_Name_Number
- Store_State

- Product_Name
- Product_Category
- Brand
- Selling_Price
- Customer_Segment/Membership
- Region

5. Feature Engineering

Several calculated fields were created to aid analysis.

a. Data Extraction

The month values were extracted from the date column using the =TEXT([@[Order_Date]],"mmm") function.

b. Total Price Value

The total price was calculated using the following formula:

Total_Price=Quantity*Unit_Price

c. Total Discount

The discounts were converted into decimals using discount/100, and the decimals were formatted as percentages. Then, the discount amount was calculated using the formula:

Discount_Amount=Total_Price*Discount_Percent

d. Net Sales Value

The total discounted price was calculated using the formula:

Total_Discounted_Price=Total_Price–Discount_Amount

e. Total Cost to Customer Including Shipping

The total cost to customer was calculated with the formula:

Total_Sales=Total_Discounted_Price+Shipping_Cost

f. Returned Items

To obtain the final total revenue the company made from sales, returned items were subtracted from the total cost to customer using the formula:

Total_Revenue=Total_Sales–Refund_Amount

g. Return Status

The return status of orders was categorized using nested IFS, AND, and OR logical statements, accounting for cancelled, pending, and completed orders:

=IFS(

[@[Order_Status]]="Cancelled", "Cancelled Order",
AND([@[Refund_Amount]]>0,OR([@[Order_Status]]="Completed",[@[Order_Status]]="Pending")), "Returned",

AND([@[Refund_Amount]]=0,
OR([@[Order_Status]]="Completed",[@[Order_Status]]="Pending")), "Not Returned"

)

h. Inventory Shortage Status

The number of inventory shortages were measured against stock reorder level using the formula:

=IFS([@[Stock_On_Hand]]>=[@[Reorder_Level]], "No Shortage",
[@[Stock_On_Hand]]<[@[Reorder_Level]],"Shortage")Interpretation:

i. Return rate

Overall return rate = Returned_Orders/Total_Orders * 100

6. EDA

Aggregate functions were used to calculate the datasets summary statistics and totals.

Metric	Total	Formula/Function

Total Revenue	₦ 6,894,815,321.25	=SUM(ShopSmart_Orders[Total_Revenue])
Total Quantity	149923	=SUM(ShopSmart_Orders[Quantity])
Avg. Selling Price	₦ 36,313.19	=AVERAGE(ShopSmart_Orders[Selling_Price])
Avg. Order Value	₦ 139,133.16	=AVERAGE(ShopSmart_Orders[Total_Sales])
Lowest Sales Amount	₦ 1,666.05	=MIN(ShopSmart_Orders[Total_Sales])
Highest Sales Amount	₦ 503,183.00	=MAX(ShopSmart_Orders[Total_Sales])
Number of Orders	50000	=COUNTA(ShopSmart_Orders[Order_ID])
Number of Customers	5000	=COUNTA(UNIQUE(ShopSmart_Orders[Customer_ID]))
Number of Products	500	=COUNTA(UNIQUE(ShopSmart_Orders[Product_ID]))
Number of Stores	25	=COUNTA(UNIQUE(ShopSmart_Orders[Store_ID]))

7. Pivot Table Analysis

Pivot Tables were used to analyze the following metrics:

Sales Analysis

- Revenue Trend per Month
- Revenue by Gender
- Revenue by Customer Membership Tiers
- Revenue by Product Category
- Revenue by Store Location
- Revenue by Region
- Revenue by State
- Revenue by Brand

- Revenue by Top 10 Customers
- Revenue by Top 10 and Bottom 10 Products

Returns Analysis

- Return Rate by Product Category – Return Rate was calculated using the formula: returned /total * 100
- Return Reason Distribution by Product Category

Inventory Analysis

- Inventory Status by Category

8. Dashboard Development

An interactive Excel dashboard was created using:

- Pivot Tables
- Pivot Charts
- Slicers and timelines
- KPI Cards

Dashboard filters included:

- Region
- Store Location
- Product Category
- Membership Tier

Pivot Charts included:

- Revenue by Month Line Chart
- Revenue by Product Category Pie Chart
- Revenue by Store Column Chart
- Revenue by State Bar Chart
- Top 10 products Bar Chart

Auxillary Charts included:

- Revenue by Region Pie Chart
- Return Rate per Category Column Chart
- Return Reason per Category Column Chart
- Shortage per Category Column Chart

- Shortage per Store Column Chart

10. Tools Used

- Microsoft Excel
- Pivot Tables
- Pivot Charts
- XLOOKUP
- IF / IFS, AND, and OR logical statements
- COUNTA
- COUNTA UNIQUE
- SUM
- MIN, MAX, and AVERAGE
- Slicers and timelines